

**PROJECT REPORT OF CO-OP Project at Industry (Module-2)**

**ON**

**“Predicting Conversions using Machine Learning”**

**submitted in partial fulfilment of the requirements for the award of degree of**

**BACHELOR OF ENGINEERING**

**In**

**COMPUTER SCIENCE AND ENGINEERING**

**Submitted by:**

**Matangini Gupta (2210994810)**

**Kritika Garg (2210004905)**

**Rachit Singh (2210994824)**

**Yashmit Gauba (2210994863)**

**Supervised By:**

**Dr. Simarjit Kaur**

**Associate Professor**

**Chitkara University**

**Institute of Engineering and Technology**



***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING***

**CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY  
CHITKARA UNIVERSITY, PUNJAB, INDIA**

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## **DECLARATION**

I hereby certify that the work which is being presented in the project report entitled “**Predicting Conversions using Machine Learning**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Simarjit Kaur**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura  
Date: 30.04.2026

**Matangini Gupta (2210994810)**  
**Kritika Garg (2210004905)**  
**Rachit Singh (2210994824)**  
**Yashmit Gauba (2210994863)**

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

**Dr. Simarjit Kaur**

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

## **ACKNOWLEDGEMENT**

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We are also thankful to the **Department of Computer Science and Engineering, Chitkara University**, for providing us with the opportunity, resources, and academic environment required to successfully complete this project. This work enhanced our understanding of machine learning models, uplift modeling, causal inference, contextual bandits, delayed feedback systems, and privacy-aware advertising technologies.

We would also like to acknowledge the support of open-source tools and technologies such as Python, Scikit-learn, XGBoost, Pandas, and Google Colab, which played an important role in the implementation of this project. Finally, we sincerely thank all team members for their dedication, teamwork, coordination, and research contributions that made this project successful.

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## **ABSTRACT**

Conversion prediction has emerged as one of the most important applications of machine learning in digital advertising and marketing analytics. Businesses use predictive systems to estimate whether users are likely to perform valuable actions such as purchasing products, subscribing to services, or installing applications after interacting with advertisements.

This project focuses on analyzing machine learning approaches for customer conversion prediction using both practical implementation and research-based analysis. The implementation project focuses on predicting whether customers will subscribe to an insurance product using demographic and marketing data. Various machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost were implemented and evaluated using AUROC metrics.

The research paper extends the problem beyond traditional supervised learning by discussing causal uplift modeling, incrementality estimation, contextual multi-armed bandits, delayed conversion feedback, and privacy-aware advertising systems. The study highlights that modern conversion optimization systems must focus not only on predictive accuracy but also on incremental business value.

The project concludes that advanced machine learning architectures combined with causal inference and adaptive optimization techniques provide more reliable and scalable conversion systems for real-world advertising environments.

# 1. INTRODUCTION

## *1.1 Overview of Conversion Prediction*

Conversion prediction is one of the most important applications of machine learning in digital marketing and advertising analytics. It refers to the process of predicting whether a customer or user will perform a desired action after interacting with a business platform, advertisement, website, or promotional campaign. These desired actions are commonly known as *conversions* and may include purchasing a product, subscribing to a service, registering on a platform, installing a mobile application, clicking on an advertisement, or completing a form submission. Machine learning algorithms analyze this data to identify hidden behavioral patterns and estimate the probability of future customer actions.

Conversion prediction systems are widely used across industries such as:

- Digital advertising
- E-commerce
- Banking and finance
- Insurance marketing
- Customer relationship management (CRM)

The importance of conversion prediction has increased significantly because modern businesses rely heavily on data-driven decision-making. Accurate conversion prediction helps organizations optimize marketing performance and improve operational efficiency. Some major applications include:

- Audience targeting and segmentation
- Advertisement ranking and recommendation
- Budget allocation across campaigns
- Customer retention strategies
- Personalized marketing
- Revenue forecasting

Traditional marketing approaches relied heavily on manual analysis and generic customer targeting. However, machine learning-based conversion prediction systems provide automated and intelligent decision-making capabilities by processing large-scale structured and behavioral datasets efficiently.

Modern machine learning techniques such as Logistic Regression, Decision Trees, Random Forest, XGBoost, Deep Neural Networks, DeepFM, and Deep Interest Networks (DIN) are capable of learning complex relationships between customer features and conversion behavior. These techniques enable businesses to improve prediction accuracy and maximize return on investment (ROI).

In this project, conversion prediction is studied from both a practical implementation perspective and a research-oriented perspective. The implementation focuses on customer conversion classification using historical insurance marketing datasets, while the research component explores advanced concepts such as uplift modeling, incrementality estimation, causal inference, contextual bandits, delayed feedback modeling, and privacy-aware advertising systems.

## 1.2 Background of the Problem

Traditional digital advertising and marketing systems primarily focus on *attributed conversions* rather than *incremental conversions*. In simple terms, most conventional systems attempt to predict whether a user will convert, but they fail to determine whether the advertisement itself actually influenced the user's decision to convert. This creates a major limitation in real-world advertising systems because many users may already have a high probability of converting even without being exposed to advertisements. As a result, businesses often spend advertising budgets on customers who would have converted naturally, leading to inefficient resource utilization.

Several practical challenges arise from this problem:

- ***Inefficient Ad Spending***  
Marketing budgets are often allocated toward users who already intend to purchase products or services. This reduces advertising efficiency and increases unnecessary expenditure.
- ***Poor Return on Ad Spend (ROAS)***  
Since attribution-based systems do not measure actual causal impact, organizations may receive lower incremental returns despite high conversion numbers.
- ***Lack of Personalization***  
Traditional systems fail to identify persuadable customers who can actually be influenced through targeted advertising strategies.
- ***Sample-Selection Bias***  
Many conversion prediction systems are trained only on clicked advertisements rather than the full impression space. This creates biased learning patterns and inaccurate probability estimation.
- ***Delayed Conversion Feedback***  
In real-world systems, conversions may occur hours or days after advertisement exposure. Delayed feedback introduces label uncertainty and negatively affects model training.
- ***Class Imbalance***  
Conversion events are generally very rare compared to non-conversion events. This creates highly imbalanced datasets, making prediction more difficult.
- ***Privacy Restrictions***  
Modern privacy regulations and browser-level tracking restrictions have reduced access to user-level behavioral data.

To address these limitations, modern research has shifted toward more advanced approaches that combine prediction with causal reasoning and adaptive optimization techniques. The research paper associated with this project discusses several advanced methodologies, including:

- Uplift Modeling – Estimating the true incremental impact of advertisements
- Causal Inference – Understanding cause-and-effect relationships in advertising systems
- Incrementality Estimation – Measuring conversions directly influenced by advertisements
- Contextual Multi-Armed Bandits – Adaptive decision-making under uncertainty
- Delayed Feedback Modeling – Handling late-arriving conversion events
- Privacy-Aware Optimization – Designing systems under privacy-preserving environments



These modern approaches aim to improve not only predictive accuracy but also business decision quality and resource efficiency.

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## ***1.2 Problem Statement***

The primary objective of this project is to develop and analyze machine learning-based customer conversion prediction systems using historical demographic and marketing datasets. The project aims to predict whether a customer is likely to convert based on multiple input features such as customer demographics, behavioral attributes, and marketing interaction data.

Although traditional machine learning models can achieve high predictive accuracy, they often face several practical limitations when deployed in real-world advertising systems. Therefore, this project also studies the broader research challenges associated with conversion optimization and advertising analytics.

The major problems addressed in this project include:

- ✓ ***Accurate Customer Conversion Prediction***  
Building machine learning models capable of classifying customers into conversion and non-conversion categories with high reliability.
- ✓ ***Handling Class Imbalance***  
Managing highly imbalanced datasets where positive conversion events are significantly fewer than negative events.
- ✓ ***Delayed Conversion Feedback***  
Studying the impact of delayed customer actions on training and evaluation processes.
- ✓ ***Sample-Selection Bias***  
Understanding bias caused by training models on partial or clicked-user datasets.
- ✓ ***Calibration and Reliability***  
Ensuring that predicted probabilities accurately reflect real-world conversion likelihood.
- ✓ ***Incrementality and Causal Impact***  
Analyzing whether conversions occur because of advertisement exposure or independent customer intent.
- ✓ ***Adaptive Decision-Making***  
Understanding how contextual bandits can optimize ad allocation and budget utilization dynamically.
- ✓ ***Privacy-Aware Measurement***  
Studying modern advertising challenges caused by privacy-preserving technologies and restricted tracking mechanisms.

The project combines practical implementation with theoretical research analysis to provide a comprehensive understanding of modern conversion prediction systems.

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## ***1.3 Objectives of the Project***

The objectives of this project are divided into primary and secondary goals.

Primary Objectives

The major objectives of the project are:

- To study machine learning techniques used in customer conversion prediction systems.
- To analyze supervised learning algorithms for classification-based conversion prediction.
- To compare different machine learning models using performance metrics such as accuracy, precision.
- To understand the importance of feature engineering and data preprocessing in predictive modeling.
- To study advanced conversion optimization concepts such as uplift modeling and causal inference.
- To understand how machine learning improves digital advertising and marketing analytics systems.

### **Secondary Objectives**

The secondary objectives of the project include:

- To study delayed conversion feedback handling methods.
- To understand sample-selection bias in advertising datasets.
- To study privacy-preserving advertising systems and signal loss challenges.
- To understand incremental conversion estimation techniques.
- To identify research gaps and future directions in conversion optimization systems.
- To explore deployment-related challenges in real-world machine learning pipelines.

Through these objectives, the project aims to bridge the gap between theoretical machine learning research and practical advertising system implementation.

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## ***1.4 Research Questions***

The research paper associated with this project investigates several important research questions related to conversion prediction and optimization systems.

**RQ1:** What machine learning models are most commonly used in conversion prediction systems? This question focuses on identifying and analyzing commonly used machine learning algorithms such as Logistic Regression, Random Forest, XGBoost, LightGBM, DeepFM used in modern advertising systems.

**RQ2:** How do uplift modeling techniques improve incremental conversion quality compared to attribution-based systems? This question finds limitations of traditional attribution-based prediction systems and studies how causal uplift modeling improves decision-making by estimating true advertisement impact.

**RQ3:** How can contextual bandits optimize ad allocation under budget constraints? This research question explores adaptive decision-making frameworks where contextual multi-armed bandits dynamically allocate advertisements while satisfying budget and pacing.

**RQ4:** What are the major implementation challenges in privacy-aware conversion systems? This question examines practical difficulties caused by modern privacy-preserving technologies such as App Tracking Transparency (ATT), delayed signals, and limited user-level tracking.

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## ***1.5 Scope of the Project***

The scope of this project covers both practical implementation and theoretical analysis of modern customer conversion prediction systems.

The project mainly focuses on:

### **1) Supervised Machine Learning**

Implementation and evaluation of classification-based machine learning algorithms for predicting customer conversion behavior.

### **2) Customer Conversion Prediction**

Analyzing customer demographic and behavioral data to estimate conversion probabilities.

### **3) Classification Algorithms**

Studying and comparing multiple machine learning models such as:

- Logistic Regression
- Random Forest
- XGBoost
- Deep Learning Models

### **4) Uplift Modeling and Causal Inference**

Understanding incremental conversion estimation and treatment-effect modeling techniques.

### **5) Contextual Multi-Armed Bandits**

Studying adaptive decision-making systems for dynamic advertisement allocation.

### **6) Delayed Feedback Modeling**

Analyzing delayed conversion challenges and time-based feedback systems.

### **7) Privacy-Aware Advertising Systems**

Understanding how privacy restrictions affect measurement, tracking, and conversion optimization.

### **8) Model Evaluation and Calibration**

Studying reliability metrics, probability calibration, and deployment readiness.

## **2. METHODOLOGY**

### ***2.1 Research Methodology***

The methodology used in this project combines both theoretical research and practical machine learning implementation to study customer conversion prediction systems. The project integrates concepts from digital marketing analytics, supervised learning, causal inference, uplift modeling, and adaptive decision-making techniques discussed in the research paper along with implementation-based experimentation from the GitHub project.

The overall workflow includes literature review, dataset preprocessing, exploratory data analysis, machine learning model implementation, and performance evaluation. The methodology focuses not only on prediction accuracy but also on practical advertising challenges such as class imbalance, delayed feedback, sample-selection bias, and privacy-aware optimization.

### **Methodology Components**

| Component                | Description                                                            |
|--------------------------|------------------------------------------------------------------------|
| Literature Review        | Study of conversion prediction research papers and advertising systems |
| Practical Implementation | Training machine learning models on marketing datasets                 |
| Comparative Analysis     | Comparing algorithms and prediction approaches                         |
| Performance Evaluation   | Measuring model accuracy and reliability                               |

*Table 1: Components of Methodology*

## 2.2 Dataset Collection

The project uses a structured insurance marketing dataset containing historical customer and campaign-related information. The dataset is used to predict whether a customer will subscribe to an insurance policy. The collected data contains demographic, behavioral, and campaign interaction features that influence customer conversion probability.

### Dataset Features

| Feature Category        | Examples                     |
|-------------------------|------------------------------|
| Demographic Information | Age, marital status          |
| Occupation Details      | Job type                     |
| Education Information   | Qualification level          |
| Campaign Information    | Contact month, call duration |
| Previous Campaign Data  | Previous campaign outcome    |

*Table 2: Features within the dataset for comparison*

- The dataset is suitable for supervised machine learning because it contains labeled conversion outcomes for customer classification.

## 2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand data distribution, feature relationships, and customer behavior patterns before model training. The analysis helped identify important variables influencing conversion outcomes.

*Objectives of EDA:*

- Understanding feature distribution
- Identifying conversion trends
- Studying customer behavior
- Detecting correlations among variables

## 2.4 Handling Imbalanced Data

The dataset was highly imbalanced because the number of converted customers was much lower than non-converted customers.

To solve this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied.

### **SMOTE Technique**

SMOTE generates synthetic minority class samples to balance the dataset and improve model learning.

### **Advantages of SMOTE**

- Balances dataset distribution
- Improves minority class prediction
- Reduces model bias
- Enhances recall and F1-score

---

## ***2.5 Machine Learning Workflow***

The project follows a structured machine learning pipeline from raw data collection to final performance evaluation.

### **Complete Workflow**

| Step                      | Description                                 |
|---------------------------|---------------------------------------------|
| Data Collection           | Gathering marketing datasets                |
| Data Cleaning             | Removing inconsistencies and missing values |
| Exploratory Data Analysis | Understanding trends and relationships      |
| Data Balancing            | Applying SMOTE                              |
| Model Training            | Training machine learning algorithms        |
| Model Evaluation          | Measuring prediction performance            |
| Performance Comparison    | Comparing different models                  |

*Table 3. Planned workflow with each step*

The workflow combines research concepts with practical implementation techniques to build an effective customer conversion prediction system.

## **3. TOOLS AND TECHNOLOGIES**

### ***3.1 Programming Language – Python***

In this project, Python was particularly useful for handling structured marketing datasets, performing exploratory analysis, implementing machine learning algorithms, and comparing model performances efficiently.

Some major reasons for selecting Python include:

- Easy syntax and faster development
- Large community and open-source support
- Availability of advanced ML libraries
- Efficient handling of tabular datasets
- Strong integration with visualization tools

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### 3.2 Machine Learning Models Used

The project implemented multiple supervised machine learning algorithms to compare prediction performance and identify the most effective model for customer conversion prediction.

Comparison of Machine Learning Models:

| Model                      | Purpose                   | Strengths                                     | Limitations                     |
|----------------------------|---------------------------|-----------------------------------------------|---------------------------------|
| <b>Logistic Regression</b> | Baseline classification   | Simple, interpretable, fast                   | Limited non-linear learning     |
| <b>Random Forest</b>       | Ensemble learning         | Handles complex patterns, reduces overfitting | Higher computational cost       |
| <b>XGBoost</b>             | Boosted ensemble learning | High accuracy, efficient, robust              | Requires tuning and computation |

Table 4. Comparing different Machine Learning models

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1) **Logistic Regression:** Logistic Regression was used as the baseline classification model in the project. It predicts the probability of customer conversion using a logistic function and is commonly applied in binary classification problems. The model is highly interpretable because it explains how different features influence conversion probability.

Advantages

- Simple and easy to implement
- Fast training and prediction
- Good interpretability
- Effective for baseline comparison

Limitations

- Cannot efficiently capture complex non-linear relationships
- Performance decreases with highly complicated feature interactions

In the research paper, Logistic Regression is discussed as a classical supervised learning approach widely used in early conversion prediction systems because of its scalability and stable probability outputs.

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2) **Random Forest:** Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction performance and reduce overfitting. In this project, Random Forest helped model complex customer behavior patterns and non-linear relationships between demographic and marketing variables.

Advantages

- Handles non-linear feature interactions
- Robust against overfitting
- Performs well on structured datasets
- Works efficiently with mixed feature types

Limitations

- Increased computational complexity
- Larger memory usage

- Less interpretable compared to Logistic Regression

The research paper highlights tree-based ensemble models as strong performers in advertising and conversion prediction systems due to their ability to capture complex behavioral relationships.

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**3) XGBoost:** XGBoost (Extreme Gradient Boosting) was the best-performing model in the implementation project. It is an advanced boosting algorithm that sequentially improves weak learners to produce highly accurate predictions.

Advantages

- High predictive accuracy
- Strong AUROC performance
- Efficient handling of sparse data
- Excellent performance on tabular datasets
- Supports regularization to reduce overfitting

Limitations

- Requires hyperparameter tuning
- Higher computational requirements
- More complex implementation

According to the research paper, boosting-based approaches such as XGBoost are highly effective for large-scale conversion prediction systems because they provide strong predictive power while handling structured advertising datasets efficiently.

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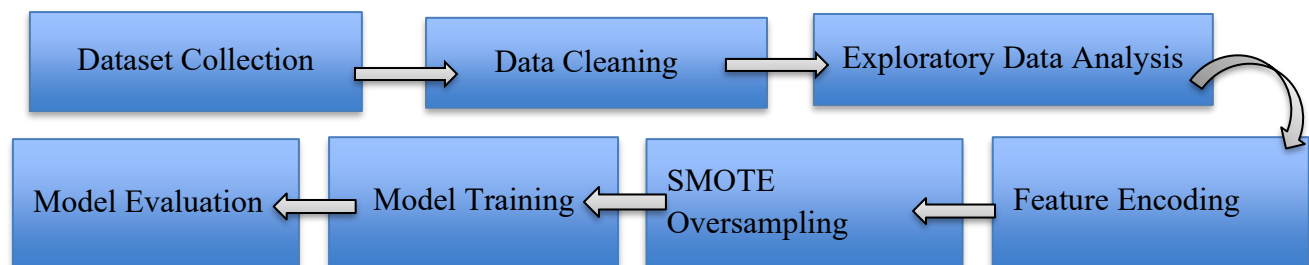
## **4. IMPLEMENTATION**

### ***4.1 System Architecture***

The implementation of the customer conversion prediction system followed a structured machine learning pipeline beginning from raw dataset collection to final model evaluation. The architecture was designed to ensure proper preprocessing, balanced learning, accurate prediction, and reliable performance comparison among different machine learning algorithms.

The project workflow combined both data engineering and machine learning stages to build an efficient customer conversion classification system.

System Flowchart:



*Figure 1. Procedural Flowchart of system architecture*

After training, all machine learning models were evaluated using multiple classification metrics to measure prediction quality and reliability.

### **Evaluation Metrics Used:**

| Metric           | Purpose                                |
|------------------|----------------------------------------|
| Accuracy         | Overall prediction correctness         |
| Precision        | Correct positive predictions           |
| Recall           | Ability to identify actual conversions |
| AUROC            | Classification quality measurement     |
| Confusion Matrix | Detailed prediction analysis           |

*Table 5. Various metric used for comparison*

### **Best Performing Model:**

| Model               | Performance Observation            |
|---------------------|------------------------------------|
| Logistic Regression | Good baseline performance          |
| Random Forest       | Improved classification capability |
| XGBoost             | Highest accuracy and AUROC         |

*Table 6. Performance based Results*

The results validated the research paper’s discussion that boosting-based ensemble methods perform exceptionally well on structured conversion prediction datasets due to their ability to capture complex feature relationships and sparse behavioral patterns.

---

## ***4.2 Research Paper Integration***

The practical implementation project mainly focused on supervised machine learning for customer conversion prediction, while the research paper expanded these concepts toward advanced advertising optimization systems. The research paper connected the implementation with modern conversion optimization approaches used in real-world digital advertising systems like:

| Concept                  | Purpose                                  |
|--------------------------|------------------------------------------|
| Uplift Modeling          | Estimate incremental conversion impact   |
| Causal Inference         | Identify true advertisement effect       |
| Contextual Bandits       | Adaptive ad allocation and optimization  |
| Delayed Feedback Systems | Handle late conversion events            |
| Privacy-Aware Systems    | Work under limited tracking environments |

*Table 7. Modern Conversion Optimization Approaches*



## 5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

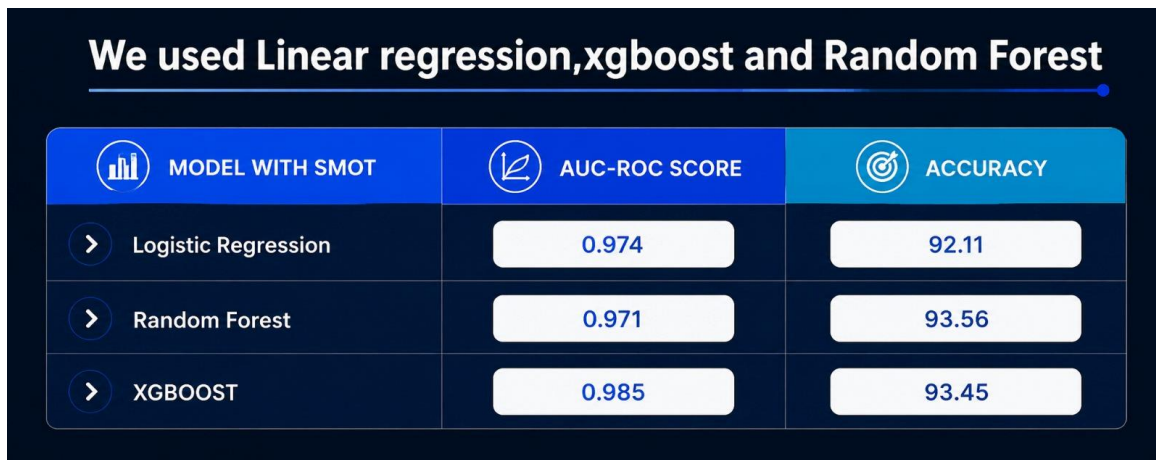
### 5.1 Performance Comparison

The implementation project compared multiple supervised machine learning models to identify the most effective approach for customer conversion prediction. The models were evaluated using classification metrics such as Accuracy, Precision, Recall and AUROC.

#### **Model Performance Comparison:**

| Model               | Performance Level | Key Observation                                |
|---------------------|-------------------|------------------------------------------------|
| Logistic Regression | Moderate          | Good baseline model with simple implementation |
| Random Forest       | High              | Better handling of non-linear relationships    |
| XGBoost             | Highest           | Best overall predictive performance            |

*Table 8. Lowest to highest performance level models*



*Fig 2. Comparison Results of all three models*

- The comparative analysis showed that ensemble-based learning models significantly outperformed traditional linear models because customer conversion behavior involves complex feature interactions and non-linear patterns.

### 5.2 Important Findings

#### 1) XGBoost Outperformed Other Models:

One of the most significant outcomes of the project was that XGBoost achieved the highest prediction accuracy and strongest AUROC performance among all implemented models.

The model performed exceptionally well because it:

- Efficiently handled structured marketing datasets
  - Learned complex feature interactions
  - Reduced prediction errors iteratively through boosting
  - Managed class imbalance more effectively
- This finding directly supports the research paper, which identifies boosting-based models as highly effective approaches for large-scale conversion prediction systems.

## 2) *Duration of Call Was the Most Important Feature:*

Exploratory Data Analysis and model training showed that call duration was one of the strongest predictors of customer conversion. Customers with longer interaction durations were more likely to subscribe to the insurance policy.

- The research paper also emphasizes that behavioral interaction features are highly valuable in modern conversion prediction systems because they represent customer intent more effectively than simple demographic information.

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## 3) *Data Imbalance Significantly Affected Performance*

The dataset used in the implementation contained a highly imbalanced class distribution, where converted customers represented only a small percentage of the total records. Without imbalance handling, machine learning models tended to favor the majority class and produced biased predictions. To solve this issue SMOTE oversampling was implemented

Effect of SMOTE:

| Before SMOTE             | After SMOTE                      |
|--------------------------|----------------------------------|
| Poor minority prediction | Improved minority classification |
| Biased model learning    | Balanced learning                |
| Lower recall             | Better recall and reliability    |

*Table 9. SMOTE Technique before and after effects*

This finding aligns with the research paper’s discussion on class imbalance and conversion labels being major challenges in real-world advertising systems.

---

## 4) *Attribution Alone is Insufficient*

One of the major conclusions from the research paper was that traditional attribution-based prediction systems are often insufficient for real business optimization. A model may predict that a customer will convert, but this does not necessarily mean that the advertisement itself caused the conversion.

The research paper explains that:

- Many users may convert naturally without advertisement influence
- Attribution models may target “sure converters”
- Businesses may waste advertising budgets inefficiently

| Traditional Prediction        | Uplift-Based Prediction          |
|-------------------------------|----------------------------------|
| Predicts observed conversions | Predicts incremental impact      |
| Focuses on attribution        | Focuses on causality             |
| May waste ad budget           | Improves optimization efficiency |

*Table 10. Traditional Vs Modern Prediction models*

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## 5) *Privacy Constraints Are Changing Advertising Systems*

Another important finding from the research paper is that modern digital advertising systems are increasingly affected by privacy restrictions and reduced user-level tracking. New privacy policies and browser restrictions limit the amount of customer behavioral data available for prediction systems.

Modern systems must therefore adapt to:

- Limited tracking data
- Privacy-preserving APIs
- Reduced third-party cookies

---

### 5.3 Comparative Analysis of Approaches

The project and research paper together compared different conversion optimization approaches used in advertising and marketing analytics systems.

| Approach             | Major Advantage                  | Limitation                      |
|----------------------|----------------------------------|---------------------------------|
| Attribution Models   | Easy implementation              | Ignores causality               |
| Predictive ML Models | High predictive accuracy         | May target sure converters      |
| Uplift Models        | Measures true incremental impact | Requires treatment-control data |
| Contextual Bandits   | Adaptive optimization            | Complex implementation          |

*Table 11. Different Approaches with Pros and Cons*

---

#### Attribution Models

Traditional attribution systems are easy to implement and widely used in advertising platforms. However, they only measure observed conversions and cannot determine whether the advertisement actually caused the customer action.

#### Predictive Machine Learning Models

Machine learning models such as Logistic Regression, Random Forest, and XGBoost improve prediction accuracy significantly by learning customer behavior patterns from historical data.

#### Uplift Modeling

Uplift modeling extends traditional prediction by estimating incremental impact instead of simple conversion probability. This approach improves advertising efficiency by identifying persuadable customers rather than users who would convert naturally.

#### Contextual Bandits

The research paper further discusses contextual bandits as adaptive decision-making systems that dynamically optimize advertisement allocation, spending, and campaign performance under changing environments and budget constraints.

## **6. CONCLUSION AND FUTURE SCOPE**

### **6.1 Conclusion**

This project successfully studied customer conversion prediction using both practical machine learning implementation and research-based analysis. The implementation focused on predicting whether customers would subscribe to an insurance policy using demographic and marketing-related data.

The complete workflow included:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- SMOTE oversampling
- Feature scaling
- Model training and evaluation

Three machine learning models were implemented and compared:

| Model                      | Observation                                                |
|----------------------------|------------------------------------------------------------|
| <b>Logistic Regression</b> | Simple and interpretable baseline model                    |
| <b>Random Forest</b>       | Better non-linear learning and reduced overfitting         |
| <b>XGBoost</b>             | Best overall performance and highest prediction capability |

*Table 12. End Result and observation based on it*

Among all models, **XGBoost** achieved the best results because of:

- Efficient boosting mechanism
  - Better handling of non-linear relationships
  - Strong feature interaction learning
  - Improved performance on structured marketing data
- ❖ Along with implementation, the research paper provided understanding of advanced conversion optimization concepts used in modern advertising systems. The study showed that traditional attribution-based systems are limited because they only predict conversions and do not measure actual incremental impact caused by advertisements.

The research emphasized the importance of:

- Uplift modeling
- Causal inference
- Incrementality estimation
- Delayed feedback handling
- Contextual multi-armed bandits
- Privacy-aware advertising systems

The paper also explained how modern advertising platforms are moving toward adaptive and intelligent optimization systems that focus not only on prediction accuracy but also on business decision quality and return on ad spend (ROAS). Overall, this project provided both practical and theoretical understanding of modern customer conversion prediction systems and their real-world applications in digital marketing and advertising analytics.

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## **6.2 Future Scope**

The future scope of this project is broad because conversion prediction systems are continuously evolving with advancements in machine learning, deep learning, causal inference, and privacy-preserving technologies.

**Future research can focus on:**

- Combining predictive models with causal inference
- Developing incrementality-aware advertising systems
- Building privacy-conscious conversion systems
- Improving delayed conversion handling
- Designing adaptive optimization systems under budget and CPA constraints

In the future, conversion prediction systems are expected to become more intelligent, adaptive, scalable, and privacy-aware. Integrating machine learning with causal inference and reinforcement learning can significantly improve modern advertising and marketing optimization systems.

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